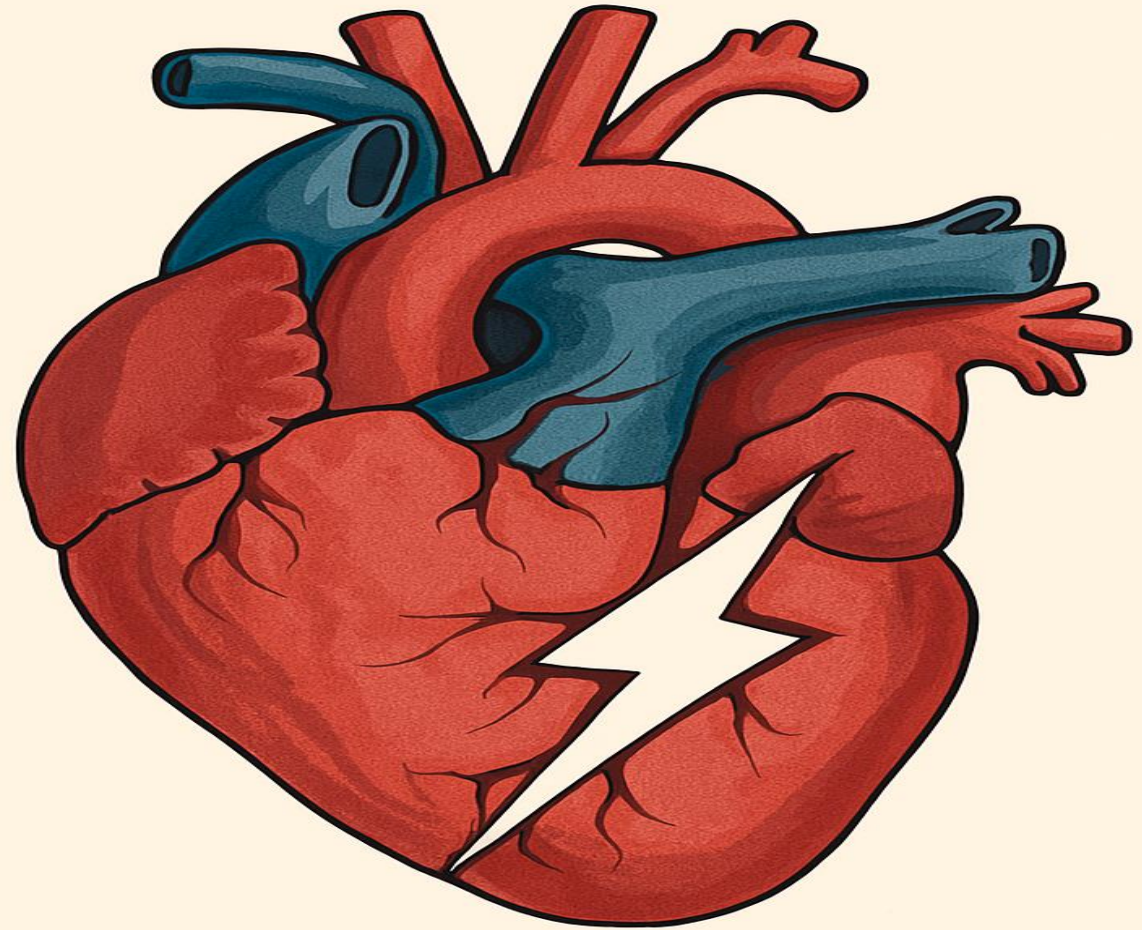


Heart Failure



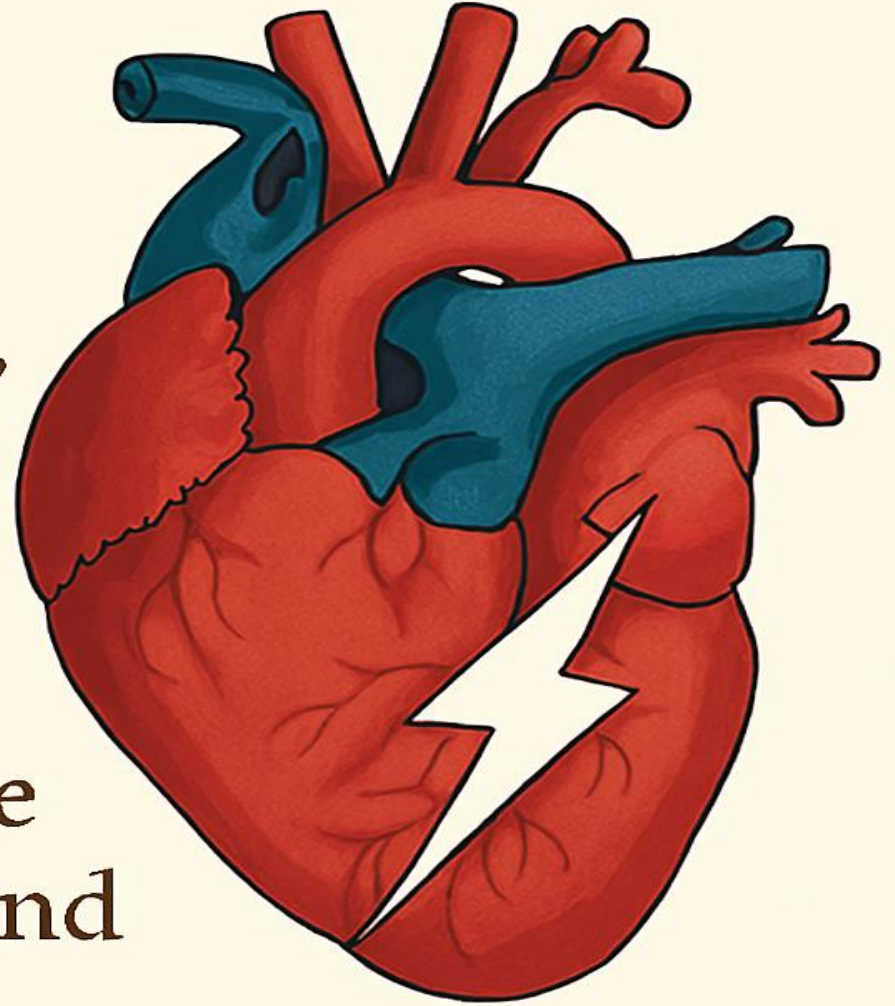
Dr/ Zinab

Agenda

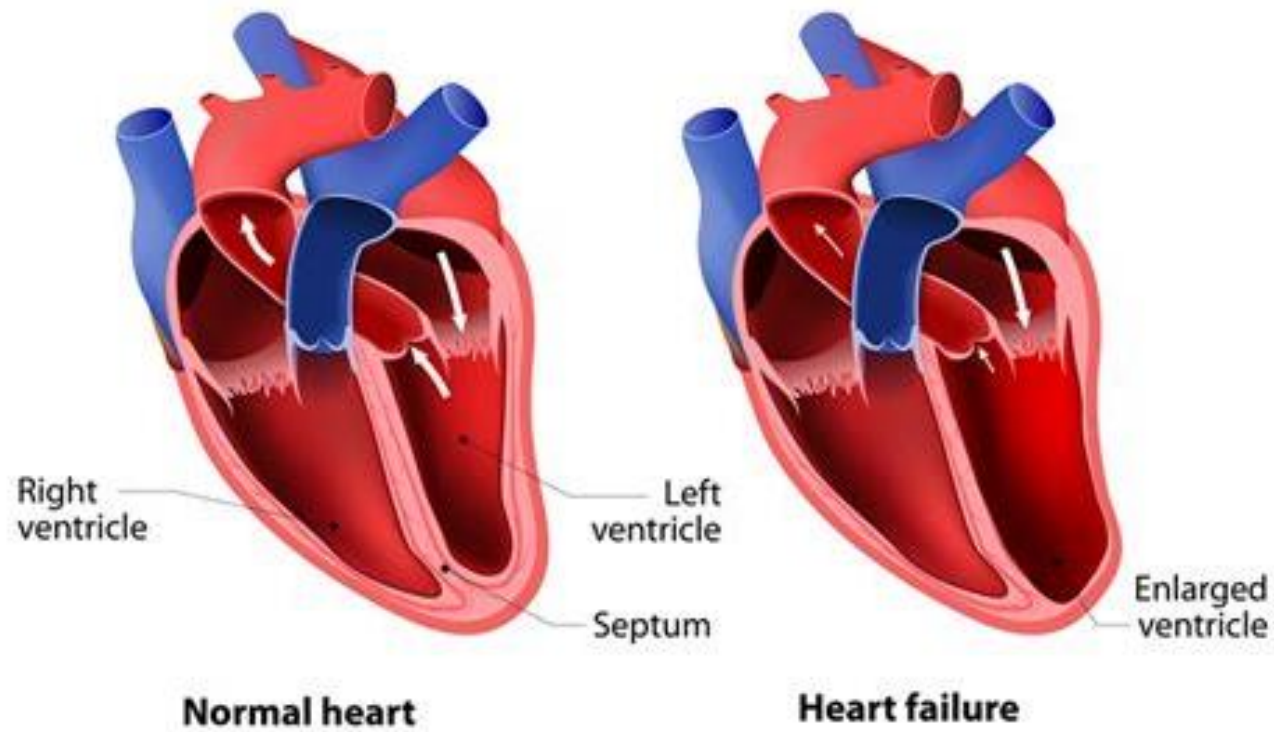
- Definition
- Causes
- Pathophysiology
- Clinical manifestation
- Diagnostic tests
- Complications
- Medical management
- Nursing management

Congestive Heart Failure (CHF)

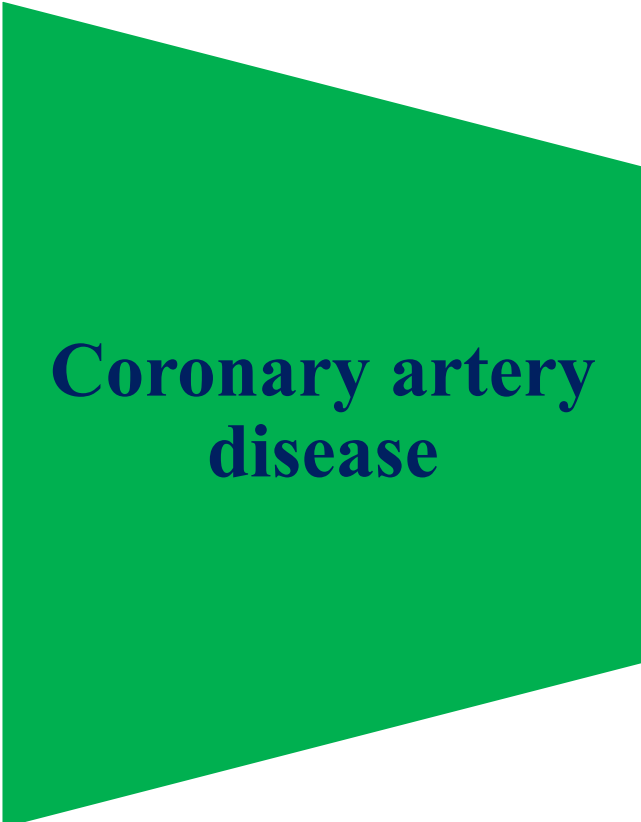
Congestive heart failure (CHF), or just heart failure, is a clinical syndrome that results from the inability of the heart to pump sufficient blood to meet the metabolic needs (oxygen and nutrients) of the body.



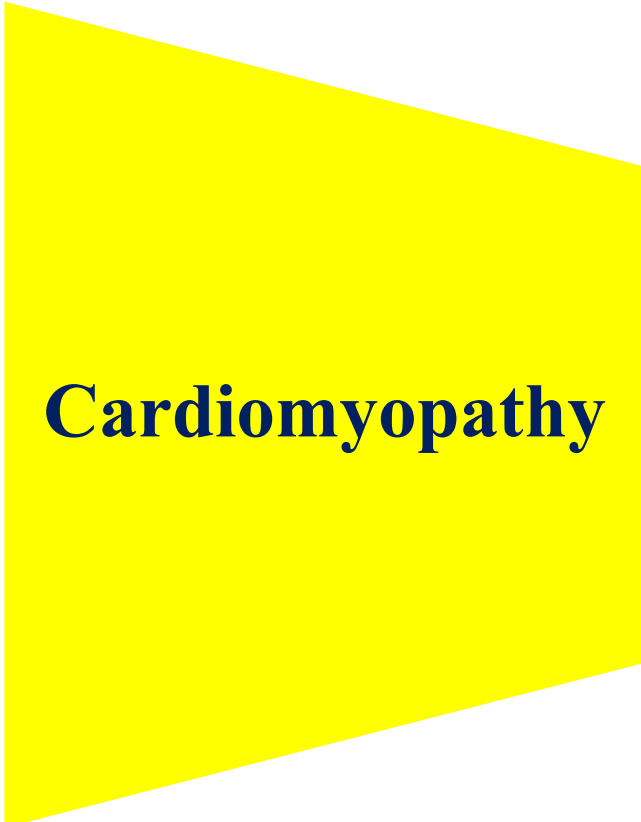
HEART FAILURE



Causes

A green parallelogram shape, tilted to the right, containing the text 'Coronary artery disease' in dark blue.

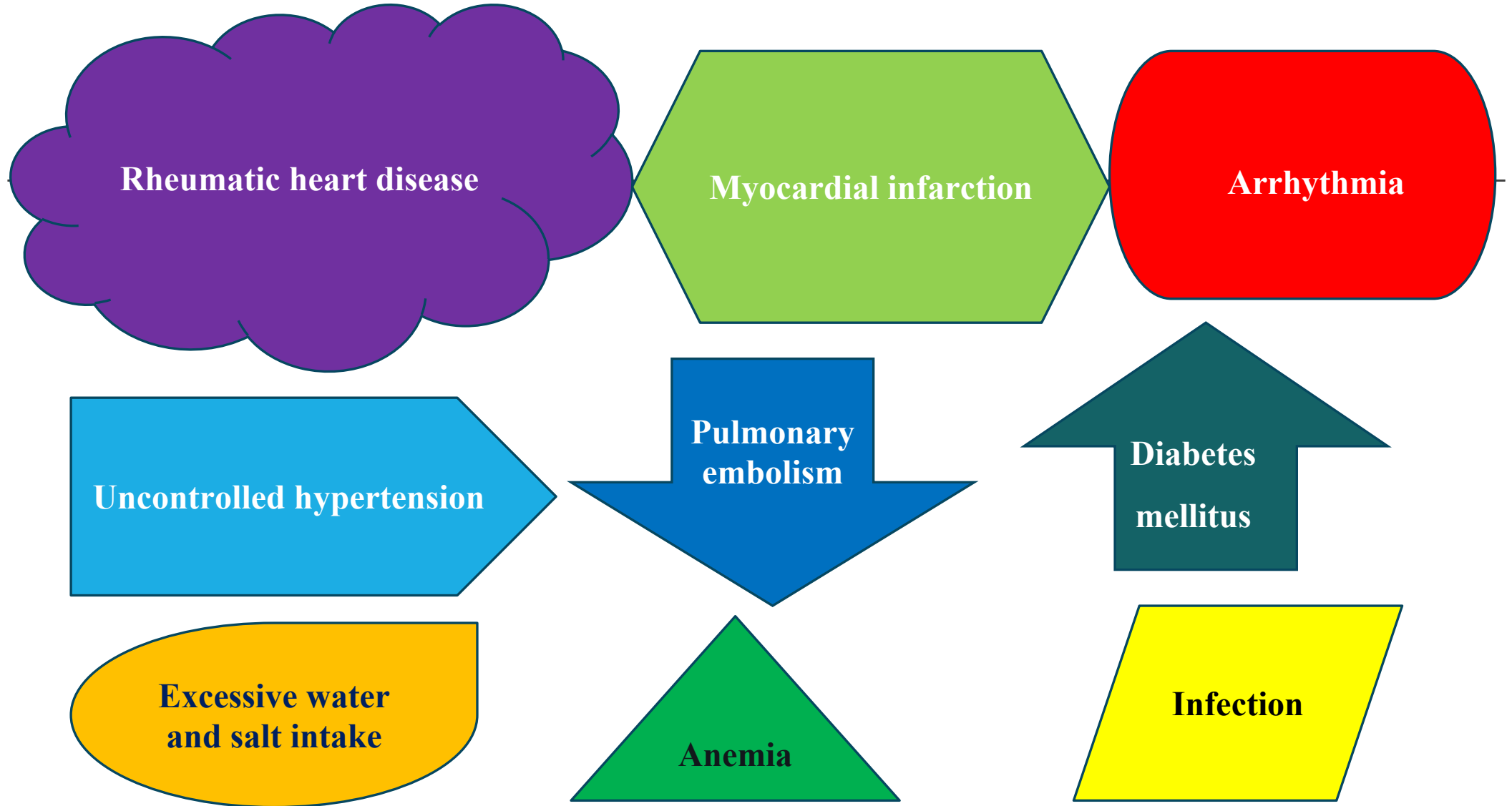
**Coronary artery
disease**

A yellow parallelogram shape, tilted to the right, containing the text 'Cardiomyopathy' in dark blue.

Cardiomyopathy

A blue parallelogram shape, tilted to the right, containing the text 'Valvular disorders' in white.

Valvular disorders



Types or Classification

1- Left-sided heart failure which may be either:

A-Systolic dysfunction

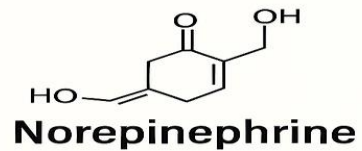
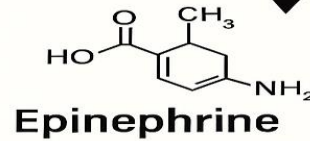
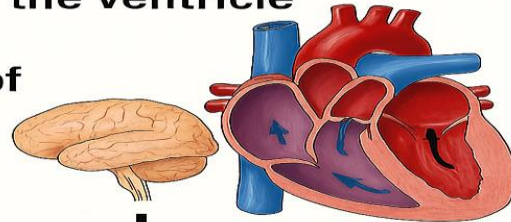
B- Diastolic dysfunction

2- Right-sided heart failure

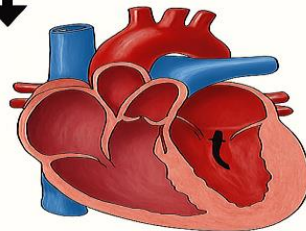
Pathophysiology of heart failure

Decrease amount of ejected blood from the ventricle

Stimulation of sympathetic nervous system

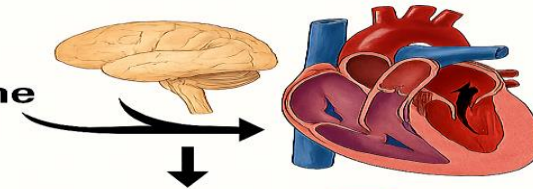


Further damage of the heart muscle occur

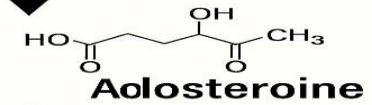
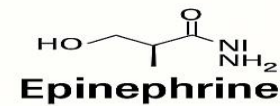


Stimulation of sympathetic nervous system along with decrease renal perfusion

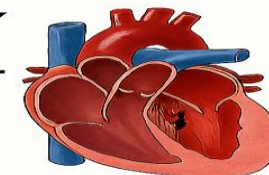
Release of rennin by the kidney



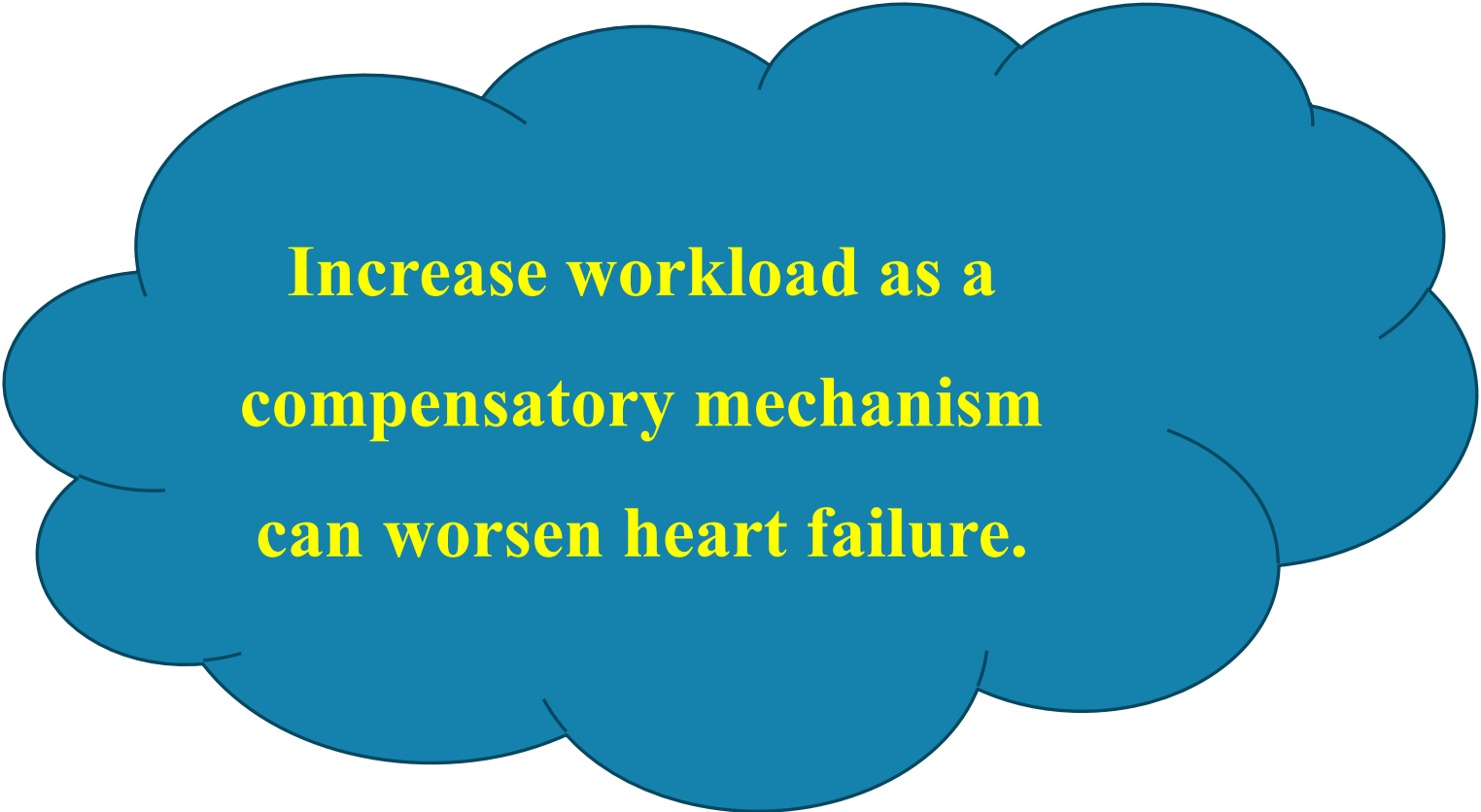
Formation of Angiotensin and Aldosterone



Increase stress on the heart and further decrease of contractility

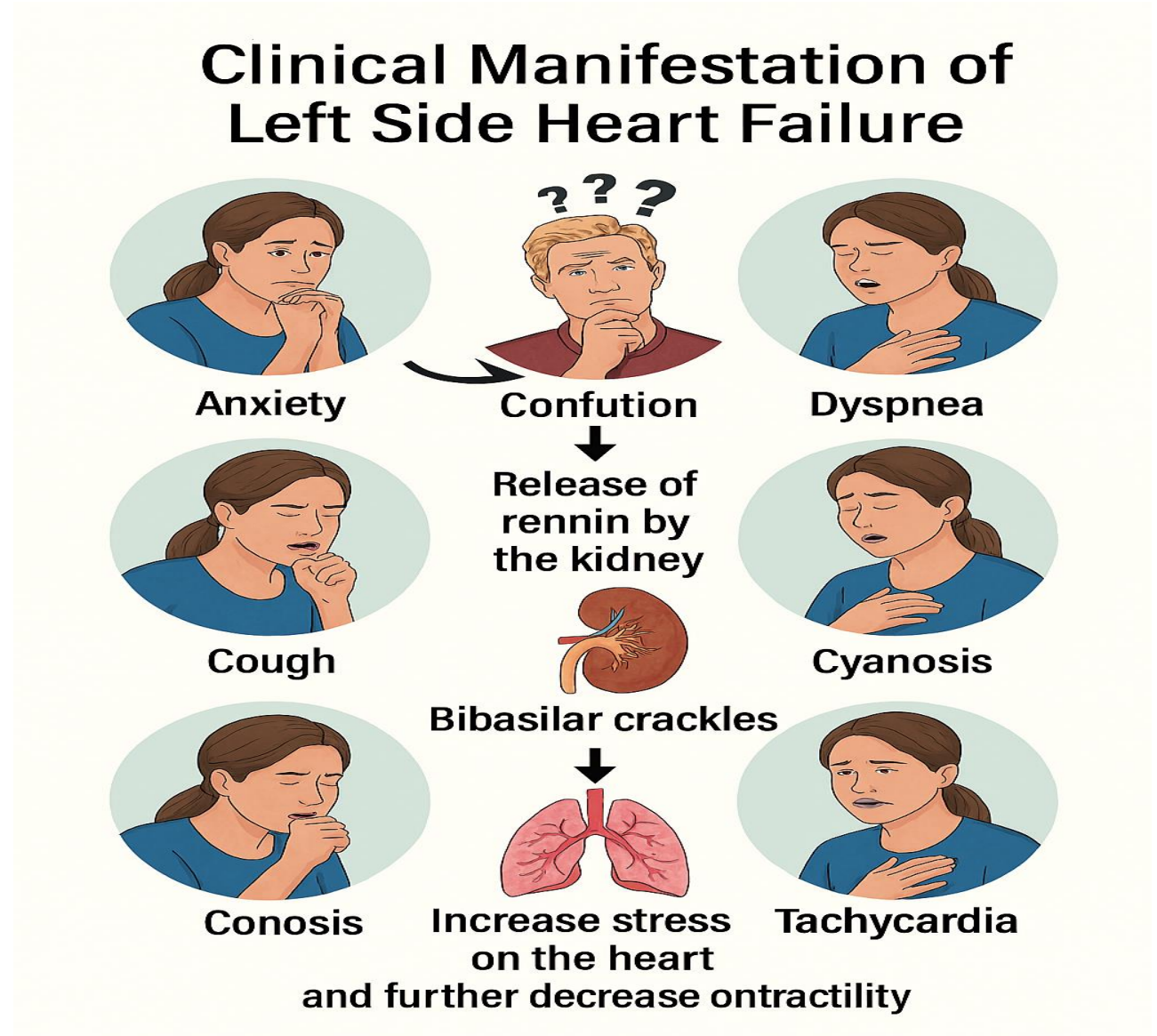


Increase heart load



**Increase workload as a
compensatory mechanism
can worsen heart failure.**

Clinical manifestation of left side heart failure



Clinical Manifestation of right-side heart failure

Clinical Manifestation of Right Side Heart Failure



Fatigue



Confusion



Dyspnea

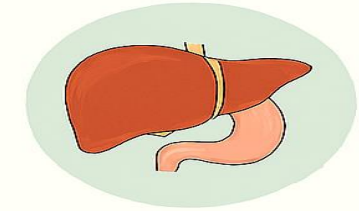


**Increased
venous pressure**

**Distended
jugular vein**



Edema



Hepatomegaly



**Increased
venous pressure**



Edema



**Anorexia and
complaints of
GI distress**

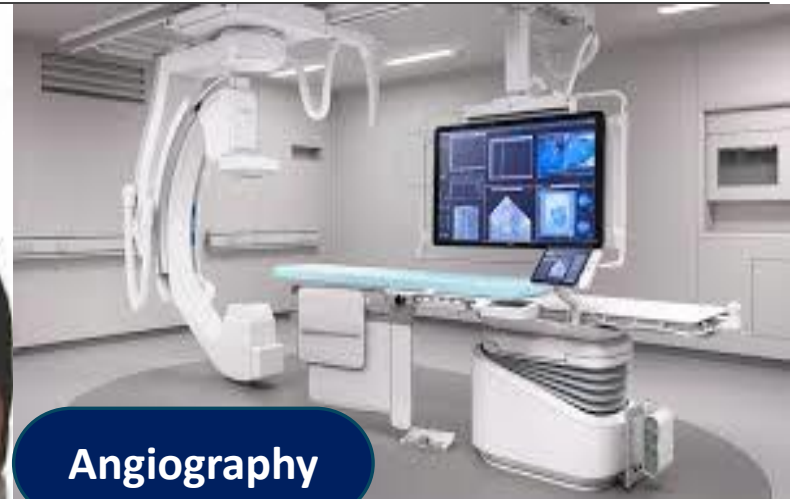
Diagnosis



Echocardiography



Chest X-ray



Angiography



Blood tests

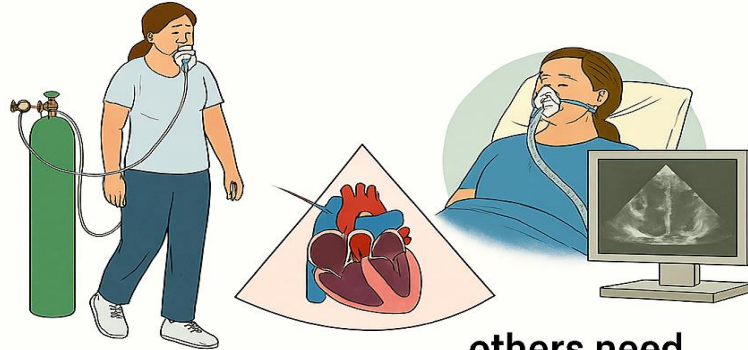
Medical Management

Treatment of CHF aims to relieve symptoms, maintain (normal fluid level in the circulatory system)

Improving prognosis by delaying progression of heart failure

Reducing cardiovascular risk.

Oxygen therapy:



Some patient
require it during
activity

others need
hospitalization for
endotracheal
intubation

Angiotensin converting enzyme

inhibitors (ACEI):
is recommended for all
patients with systolic
heart failure such as
Captopril (Capoten).



Beta-blockers:
can reduce blood pressure
and slow a rapid heart
rhythm such as
(Inderal)



Diuretics: to increase the rate of urine production and the removal of excessive extracellular fluids. The most common types are:

- **Thiazide diuretic**
- **Loop diuretic**
- **Potassium sparing diuretics.**

Thiazide diuretics as Zaroxolyn, which increases potassium and bicarbonate excretion. It used with mild fluid overload



Loop diuretics as Lasix is used with more severe fluid overload or with renal insufficiency.



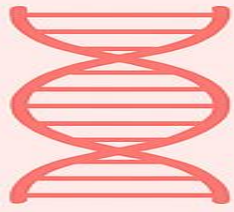
Spironalactone as Aldactone that inhabits sodium reabsorption in the late distal tubule and collecting duct.



Digitalis or digoxin.

- It increases the force of myocardial contraction.
- It enhances diuresis to relieve edema.





DIGITALIS TOXICITY

It occurs when serum level of Digoxin increases but it can occur spontaneously.

EARLY EFFECTS

- Fatigue
- Depression
- Malaise
- Anorexia
- Nausea and vomiting

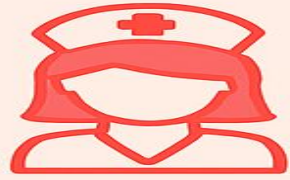


Atrioventricular block

TO REVERSE

- Hold the medication.
- If toxicity is severe, if Digoxin immune (Digibind) may be prescribed to bind with digitalis.





DIGITALIS

NURSING INTERVENTION

- **Assess** patient response to Digoxin therapy by relating signs and symptoms
- **Monitor for GI** side effects as nausea, vomiting, or abdominal pain
- **Monitor for neurological** side effects as headache, or malaise
- **Monitor** serum **potassium** level in patient receiving Digoxin because an uncorrected Potassium imbalance predisposes patient to Digoxin toxicity
- **Monitor** patient for factors that increase the risk of toxicity as:
 1. Oral antibiotics
 2. Hypokalemia
 3. Impaired renal function.
- **Assess** apical heart rate before administering Digoxin. If it very rapid or below 60 beats, withhold the medication and notify the physician because these signs



Diet and lifestyle modifications

**Monitor
weight**

**Moderate
physical
activity**

**Sodium
restriction**

**Fluid
restriction**

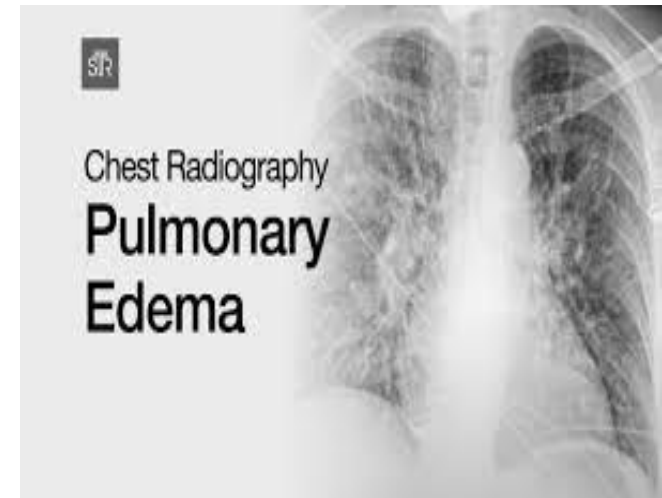
**Correction of the related
causes of heart failure e.g.
heart valve repair surgery**

Complications

a) Acute pulmonary edema

b) Cardiogenic shock

c) Refractory heart failure

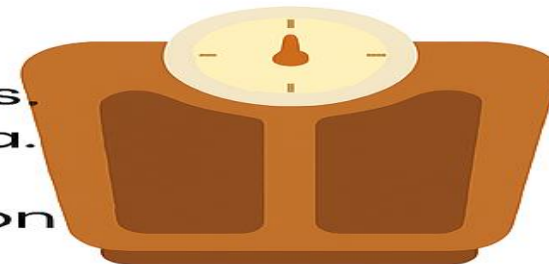


Nursing Process



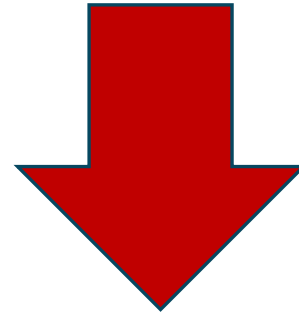
ASSESSMENT

- **Observe** the effectiveness of therapy.
- **Observe** patient ability to implement self-management.
- **Observe** signs and symptoms of fluid overload.
- **Observe** patient emotional response to the diagnosis
- **Assess** sleep disturbances due to shortness of breath.
- **Assess** number of pillow needed for sleep.
- **Assess** activities of daily living and if it causes dyspnea.
- Keeping 24 hours intake and output record
- Weight patient daily
- Auscultate lung sounds at least daily to assess crackles.
- Assess the severity of edema.
- Monitor hypotension and dehydration as a complication of diuretics.



Nursing Diagnosis

Planning



1- Activity intolerance related to imbalance between oxygen supply and demand.

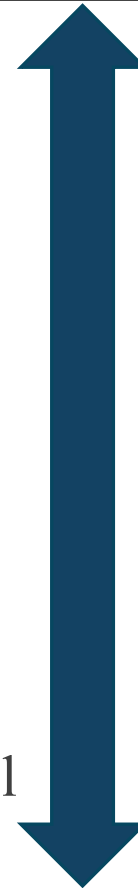
2- Excess fluid volume related to excessive fluid/sodium intake.

3- Anxiety related to decrease oxygen level

1-Promote activity

2- Relieve fluid overload

3- Decrease incidence of anxiety



Implementation

Avoid activities
outside in extreme
weather (hot or cold)

Encourage patient to
perform an activity
more slowly

Wait 2 hours after
eating to perform
activities

Stop activities with
dyspnea or pain

Measure vital signs
especially pulse
before, during and
immediately after
activity

**Oral diuretics should
be administered
early in the morning**

**Monitor the patient
fluid volume status**

**Monitor intake and
output**

**Adhere to low sodium
diet**

Patient Teaching

- Obtain adequate rest
- Check own pulse rate daily
- Accept restricting sodium intake may be permanent
- Keep regular appointment with physician or clinic

Report immediately to the physician any of the following:

a- weight gain

b-loss of appetite

c- Shortness of breath with activities

d- Swelling of the ankles, feet or abdomen

E- Persistent cough

Thank
You